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BABA INSTITUTE OF TECHNOLOGY AND SCIENCES

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Bakkannapalem Village, Madhurawada Post, Visakhapatnam - 530 048

Date: 20-12-2017

As per the discussion with management it is decided that a faculty financial support of Rs. 500 is sanctioned towards ICMSEA-2017 paper publication fee for the following faculty members.

S.No	Name of the Faculty Member	S.No	Name of the Faculty Member	S. No	Name of the Faculty Member
1	M.Pawan Kumar	11	M.Chiranjeevi	21	K.Pradeep
2	N.Sai Kumar	12	A.Uday Kumar	22	Mr. K.S.N. Murthy
3	P.Padma Rao	13	Vikas Ranjan	23	S.Kavitha
4	D.Narasimha Murty	14	D.V.Padmaja	24	P.B.Nagendra Kiran
5	Dr.N.Victor Babu	15	P.Surya Rao	25	Dr. P. Sanjeevi
6	M.Vasudeva Rao	16	Ch.Uma Chaitanya	26	K.Sireesha
7	Ch.Grace Evangelin	17	Vikas Ranjan	27	Dr.B.Rama Jyothi
8	N.S.Sindhuja	18	Hapsiba Seeli	28	M. Hima Bindu
9	M.Mohan	19	N.Swami		
10	B.V.R Murthy	20	B.Gowri		

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Paper ID: 1032

Exquisite Explanation for the Recognized Conjecture on Fermat Pseudoprimes in Square Form

Narasimha murty Dusi

Assistant Professor, Department of Mathematics,

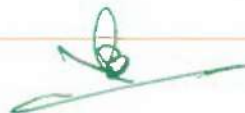
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Abstract

In this paper we will solve conjecture posed by M.Coman on primes by giving suitable example. We will bring a generalized theorem for square primes, which will deal with the generalization of Fermat Pseudoprime, which gives infinitely many pseudoprimes for different base system.

Keywords: Fermat little theorem; Fermat pseudoprime; Weifrigh primes;

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Paper ID: 2001

Effect on Compressive Strength of Concrete by Substitution Of Fine Aggregate With Blast Furnace Slag

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Dr.N.Victor Babu, Professor, Department of Civil Engineering, BABA Institute of Technology & Sciences, Visakhapatnam.

Abstract

The iron industries produce a huge quantity of blast furnace slag as by-product, which is a non-biodegradable waste material from that only a small percentage of it is used by cement industries to manufacture cement. In present inspection Blast Furnace Slag from Vizag-Steel Plant has been utilized for finding its appropriateness as a Fine Aggregate in concrete. Replacement of some natural aggregates with Slag would lead to abundant environmental benefits. It is interesting that the Blast Furnace Slag (BFS) can be used as a fine aggregate in concrete. Based on overall experimental results, it could be as- s- able that the Slag could be effectively utilized as fine aggregate in concrete practice.

Keywords: Blast Furnace Slag (BFS), Fine Aggregate; Concrete; Environmental benefits; Compressive Strength; Slag Aggregate; Iron Industries;


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Paper ID:2004

Study of normalized difference built-up (NDBI) index in automatically mapping urban areas from Landsat TM imagery.

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Abstract

The Visakhapatnam city lies on the eastern part of India and the urban sprawl in Visakhapatnam town was analyzed using multi-temporal Landsat TM data from 2005 to 2015. Spectral indices namely Normalized Difference Vegetation Index (NDVI), Normalized Difference Built-up Index (NDBI) and Built-up Index (BUI) were generated from the Landsat TM bands covering visible Red (R), Near Infrared (NIR) and Short Wave Infrared (SWIR) wave-length regions. Spectral variations in built-up, open spaces, urban vegetation and water areas were studied by generating two-dimensional spectral plots of NDBI and BUI. Remote sensing images are mainly used for monitoring and detecting the land cover changes

that occur frequently in urban and sub-urban areas as a consequence of incessant urbanization. The conversion of satellite imagery into land cover map using the existing methods of manual interpretation and parametric image classification digitally is a lengthy process. Normalized Difference Built-up Index (NDBI) which is used for mapping built-up areas automatically. This method main advantage is the unique spectral response of built-up area and other land covers. Built-up area can be effectively mapped through arithmetic manipulation of re-coded Normalized Difference Vegetation Index (NDVI) and NDBI images derived from TM imagery. This NDBI method is applied to Vishakaptanam Area in this paper and the mapped area results at accuracy of 93.9% indicates that it can be used to fulfill the mapping objective reliably and efficiently. Comparing NDBI with the maximum likelihood classification method, this NDBI method is able to serve as a worthwhile alternative for quickly and objectively mapping built-up areas.

Key words: Difference Vegetation Index, Normalized Difference Built-up Index, Built-up Index and Spectral variations.


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BAKKAI THADALEM (V), RM. PALEM,
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Paper ID: 2006

Soil Stabilization Using Plastics

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Abstract

It reflects that plastic wastes can be used in stabilization of soil which is concluded from various tests conducted on fiber reinforced soil with varying fiber content. In this study, different means of plastic waste as shopping bags which are locally available are used so as a reinforcement to perform unconfined compressive test while mixing with soil for improving engineering performance of sub grade soil. In this the plastic strips which are collected for stabilization of soil were mixed randomly with the soil. With this a series of tests were carried out on randomly reinforced soil with varying percentage of plastic strips starting from 0, 0.5%, 1.5% and 2.5%. Use of plastic in soil in an appropriate amount really aids in improving the strength of soil and can be useful as sub grade soil which can be used in pavement construction.

Keywords: Stabilization; Unconfined Compressive Strength; Fiber Reinforced Soil; Soil Properties; Shear Strength.



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2nd International Conference on Mathematical Sciences in Engineering Applications (ICMSEA - 2017)

Paper ID: 2008

WATERSHED PRESERVATION AND CONTROL USING GIS

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Abstract

Remote sensing (RS) and Geographical Information System (GIS) data provides timely, accurate and reliable information at definite intervals in a cost effective manner and these tools are helpful in mapping and monitoring of natural resources available for proper utilization. The Meghadrigedda watershed is selected for the present study and has a gross capacity of 1169 Mc. Ft. For the identification of morphological features and analyzing their properties of the Meghadrigedda basin various GIS and image processing techniques have been adopted. The linear and aerial aspects of the basin were calculated and computed. It is 7th order drainage basin and drainage pattern is dendritic type. The results obtained from watershed delineation and prioritization has wider application in preservation of the watershed.

Keywords: GIS; Watershed; Delineation; Drainage Parameters; Preservation.


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International Conference on Mathematical Sciences in Engineering Applications (ICMSEA - 2017)

Paper ID: 2008

WATERSHED PRESERVATION AND CONTROL USING GIS

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Abstract

Remote sensing (RS) and Geographical Information System (GIS) data provides timely, accurate and reliable information at definite intervals in a cost effective manner and these tools are helpful in mapping and monitoring of natural resources available for proper utilization. The Meghadrigedda watershed is selected for the present study and has a gross capacity of 1169 Mc. Ft. For the identification of morphological features and analyzing their properties of the Meghadrigedda basin various GIS and image processing techniques have been adopted. The linear and aerial aspects of the basin were calculated and computed. It is 7th order drainage basin and drainage pattern is dendritic type. The results obtained from watershed delineation and prioritization has wider application in preservation of the watershed.

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WATERSHED PRESERVATION AND CONTROL USING GIS

V.Naga Sai Sindhuja*

Dr. N. Victor Babu**

CH.Ramesh Naidu***

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Abstract

Remote sensing (RS) and Geographical Information System (GIS) data provides timely, accurate and reliable information at definite intervals in a cost effective manner and these tools are helpful in mapping and monitoring of natural resources available for proper utilization. The Meghadrigedda watershed is selected for the present study and has a gross capacity of 1169 Mc. Ft. For the identification of morphological features and analyzing their properties of the Meghadrigedda basin various GIS and image processing techniques have been adopted. The linear and aerial aspects of the basin were calculated and computed. It is 7th order drainage basin and drainage pattern is dendritic type. The results obtained from watershed delineation and prioritization has wider application in preservation of the watershed.

Keywords:

GIS;
Watershed;
Delineation;
Drainage Parameters;
Preservation.

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1.Introduction

Basin morphometry is the mathematical quantification of various drainage basin characteristics. Land is one of the important natural resource and it should be conserved to mitigate the increasing demand of land and water resources (Panhalkar S.S et al, 2012). In India 70% of the people depend on agriculture directly or indirectly. The analysis of the drainage basin is important in assessing the ground water potential and ground water management. The study on morphometric analysis using remote sensing and GIS techniques was carried out by (Sangita Mishra et al, (2010); Biswas et al. 1999; Panhalkar S.S et al, 2012; Subodh Chandra Pal and Gopal Chandra Debnath, 2012; Jagadeeswara Rao, Harikrishna P and Suryaprakasa Rao.B, 2006).

The Meghadrigedda watershed is the area enclosed between Northern latitude and Eastern longitude, covering an area of 453.40 sq. km falling in Survey of India (SOI) toposheet Nos: 65 O/1, O/2&O/3, O/5 and O/6 on 1:50,000 scale. Meghadrigedda reservoir was formed across Meghadrigedda and Naravagedda just 400 mts above their confluence to supply drinking water to the people of Visakhapatnam city at the rate of 8 Million gallons per day. The 8MGD of water is being drawn from the reservoir and distributed to various industries and industrial colonies by Visakhapatnam Municipal Corporation.

Meghadrigedda reservoir has a gross capacity of 1169 Mc.Ft and the flood bank was formed on the left side of the reservoir to protect the Howrah-Visakhapatnam railway line and the Chintalagraharam village. The average annual rainfall is around 110cm and occurs in the months of June to November. December and January are coldest months in this region and April and May are hottest months. The area enjoys sub tropical climate and the temperature ranges from 14 to 22° C in December and 33-45°C in May. Relative humidity is high in day time and varies between 70-80% throughout the year.

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****Head, GIS Technology & Applications, Xinthe Technologies Pvt. Ltd, Visakhapatnam.

Paper ID: 2011

Phytoremediation(Plants as reactors)

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Abstract

Now -a- Days we have two types of treatment processes they are physiochemical process and biological process. In biological process ASP, trickling filters, RBC, oxidation ponds, and phytoremediation. Present phytoremediation is in the early stage of commercialization for treatment of wastewater containing organic and inorganic pollutants, and in the future it may provide a low cost option. In phytoremediation we use plants for treating waste water. In that roots are influent collectors and leaves are product deliver. In this we have six reactions takes place they are phytoextraction, rhizofiltration, phytostabilization, phytodegradation, rhizodegradation and phytovolatilization. Those plants also help prevent wind, rain and groundwater from carrying pollutants away from sites to other areas. Wetland construction will help to remove or reconciliation of wastewater to environment from hazardous pollutants. Yellow or White Water Lilies, Alfalfa grass, these type of plants are useful to remove organic pollutants. Brassica family (Indian Mustard & Broccoli), Tomato plant, sunflower these are useful to remove inorganic pollutants.

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Paper ID: 2012

Experimental investigation on partial replacement of Cement with GGBFS and Sand with Bottom Ash in concrete

B.V.Ramanamurthy, Dr.N.Victor Babu Neeli, Harikrishna, B.V.Ramanamurthy

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Abstract:

Nowadays Cement is major constituent material in construction industry and Lime and Silica produces the Cement. Without these natural materials production of cement is very tedious task. Moreover the emission of Carbondioxide(CO_2) is more in the production of Cement. It causes environmental pollution globally. Sand is also naturally available material, it gives good strength to the concrete. But environmental degradation happened by scarcity of natural sand. The main motto of this work is to find alternate materials for partial replacement of Cement and sand in concrete to obtain required strength as well as to minimize degradation of natural materials. In this work Cement is partially replaced by Ground Granulated Blast Furnace slag(GGBFS) and sand is Bottom Ash(BA) to achieve required compressive strength. By this investigation, replacement of 10% Cement and 10% sand in concrete with and without adding of admixture gave sufficient characteristic compressive strength of concrete for 28 days curing period.

Key words: Characteristic compressive strength Ground Granulated Blast Furnace slag(GGBFS) Bottom Ash(BA).

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Paper ID: 2012

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B.V.Ramanamurthy, Dr.N.Victor Babu Neeli, Harikrishna, B.V.Ramanamurthy

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Paper ID: 2013

Comparative Study On End Moments Regarding Application Of Analytical Methods And Staad.Pro Software

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Email: chirum4u@gmail.com

Abstract

Analysis of portal frames involves lot of complications and tedious calculations by conventional methods. To carry out such analysis it is a time consuming task. The Moment Distribution Method & Kani's Method for analysis of portal frames can be handy in approximate and quick analysis so as to get the detailed estimates ready. In this work, these two methods have been applied only for uniform distributed loading conditions. This paper mainly deals with the comparative analysis of the results obtained from

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COMPARATIVE STUDY ON END MOMENTS REGARDING APPLICATION OF ANALYTICAL METHODS AND STAAD.Pro SOFTWARE

Chiranjeevi.M*
Udaykumar.A**
S.V.Saikumar***

Abstract

Analysis of portal frames involves lot of complications and tedious calculations by conventional methods. To carry out such analysis it is a time consuming task. The Moment Distribution Method & Kani's Method for analysis of portal frames can be handy in approximate and quick analysis so as to get the detailed estimates ready. In this work, these two methods have been applied only for uniform distributed loading conditions. This paper mainly deals with the comparative analysis of the results obtained from the analysis of single bay portal frame when analyzed manually by using STAAD.Pro Software separately. The result obtained from manually is mostly matched with the results obtained from STAAD.Pro software.

Keywords:

Portal Frame;
Moment Distribution Method;
Kani's Method;
STAAD.Pro Software.

Author correspondence:

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Asst.Prof, Department of Civil Engg.,
Baba Institute of Technology and Sciences, P.M.Palem, Visakhapatnam

Introduction

For the analysis of portal frames, Moment distribution method and Kani's method of analysis are mainly used, which allows the engineer to analyse frames easily and design it economically. The research is concluded by evaluating a selection of portal frame, with practical dimensions in order to substantiate the conclusions as stated below.

This paper presents the analysis of portal frame, considering mainly the case of single bay, which is the most common in practice. Portal frames are very efficient and economical when used for single-storey buildings, provided that the design details are cost effective and the design parameters and assumptions are well chosen. Portal frames consists of a horizontal beam resting on two columns are single storey with pitched or flat roof (Fig.1). The junction of the beam with the column consists of rigid joints and fixed at the base. The main objective of this paper is analysing the portal frame by manually and compare their results with the help of STAAD.Pro software.

2. History and Methods of Analysis

2.1 Structure

In engineering and architecture, a structure is the combination of two or more basic structural components connected together in such a way that they serve the user functionally and carry the loads arising out of self and super-imposed loads safely without causing any problem of utility.

2.2 Structural Analysis

Structural analysis deals with study and determination of forces in various components of a structure subjected to loads. As the structural system as a whole and the loads acting on it may be of composite nature

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Technical Advancement By Rice Husk Ash In Low Cost Housing Construction

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Abstract

The improvement of concrete technology can able to decrease the pollutants on environment and use of natural resources. This paper explains the usefulness of using Rice husk ash (RHA) in concrete production as a partial replacement of cement. After replacement, this can improve the strength and durability properties of concrete. RHA has high reactivity and pozzolonic property and this property depends on temperature. Concrete mixtures were produced and tested both with RHA and without RHA replacement in terms of compressive strength and results have been compared between both of them. 30% replacement of cement with RHA has been done in all cases. Soil stabilization and ground improvement techniques have become an important issue in construction field now a days. The common soil stabilization methods are becoming more costly because of high cost of stabilizing agents like lime, cement etc. The cost of the stabilization may be decreased by proper replacement of RHA with suitable proportion. Replacement with RHA and also different tests have been conducted to check the effectiveness of RHA in the soil stabilization aspect. So that RHA usage can reduce the cost of the total construction.

Keywords: Rice husk ash (RHA); Soil stabilization; Replacement of Cement; Concrete technology; Pozzolonic



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Technical Advancement By Rice Husk Ash In Low Cost Housing Construction

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Abstract

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Keywords: Rice husk ash (RHA); Soil stabilization; Replacement of Cement; Concrete technology; Pozzolonic



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Paper ID: 2015

Use of Recycled Materials In Concrete Construction

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Abstract

Use of recycled aggregate in concrete can be useful for environmental protection. Recycled aggregate are the materials for the future. The application of recycled aggregate has been started in a large number of construction projects of many European, American, Russian and Asian countries. This paper represents the basic properties of recycled fine aggregate & also compares these properties with natural aggregate. Similarly the properties of recycled aggregate concrete are also determined. Basic concrete properties

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USE OF RECYCLED MATERIALS IN CONCRETE CONSTRUCTION

Mr.Nanubilli Ramu*
Dr.N.Victor Babu**

Abstract

Use of recycled aggregate in concrete can be useful for environmental protection. Recycled aggregates are the materials for the future. The application of recycled aggregate has been started in a large number of construction projects of many European, American, Russian and Asian countries. This paper reports the basic properties of recycled fine aggregate & also compares these properties with natural aggregates. Similarly the properties of recycled aggregate concrete are also determined. Basic concrete properties like compressive strength, split tensile strength, flexural strength and workability etc. are explained here for different combinations of recycled aggregate with natural aggregate. The concrete mixes were designed using IS 10262-2009. Fine aggregate is partially replaced with 10%, 20%, 30% and cement is partially replaced with silica fume of 0%, 10% and 15% respectively.

Keywords:

Rubber aggregate;
Normal aggregate;
Cement;
Silica fume.

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1. Introduction

Waste materials are common problems in modern living. Waste accumulates from a number of sources including domestic, industrial, commercial and construction. These waste materials have to be eventually disposed of in ways that do not endanger human health. In light of this, waste minimization is increasingly seen as an ecologically sustainable strategy for alleviating the need for the disposal of waste materials, which is often costly, time and space consuming, and can also have significant detrimental impacts on the natural environment. The use of recycled materials is often cheaper for the consumers of the end product. Hence, there is also an economic justification for promoting its use.

The use of recycled materials generated from transportation, industrial, municipal and mining processes in

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Strength Improvement of M50 grade concrete as a partial replacement of fine aggregate with Stone Crusher Dust.

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Abstract

Concrete is used widely in all types of constructions ranging from small buildings to large structures like dams or reservoirs. It is the most widely used construction material to improve the strength & quality of structural elements. Sand is one of the important ingredients in the making of concrete but the availability of sand is a tedious task. For the protection of environment, almost all local governments have to be taken a decision to minimize sand excavation for utilization in construction. The current study an attempt has been made to studying the behavior of the concrete made by partial replacement of sand with Stone Crusher Dust (SCD) using admixture. The design mix prepared for M53 grade concrete as per IS Code and fine aggregate has been replaced by Stone Crusher Dust at 10%, 20%, 30%, 40% and 50% by weight. The strength tests have been made to study for 3, 7 and 28 days curing period.

Key words: Compressive strength, Sand Crusher Dust (SCD), Admixture.

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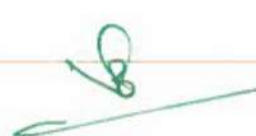
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PAPER ID: 2025

Comparative Study On End Moments Regarding Application Of Slope Deflection, Flexibility Matrix Method And Staad.Pro Software

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Abstract

In the analysis of continuous beam, there are many complexities and monotonous calculations involved in traditional methods. End moment is one of the main considerations from the structure design. In this present study determine the end moment's two various procedures i.e. Slope deflection and flexibility matrix method have been used for continuous beam. The present work shows that the End moments determined from these methods have almost same result when compared with STAAD.Pro Software. The results determined from slope deflection method are quite accurate value then the compare with the STAAD.Pro

Keywords: Continuous Beam; Slope Deflection Method; Flexibility Matrix Method; STAAD.Pro.

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Paper ID 4014

Determination of Thermophysical Properties of Nano Hybrid Refrigerants to be used in Commercial Refrigeration System

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Abstract

In present scenario, the need of increased cooling rate in commercial applications of a Refrigeration system is high. However, the complications in power consumption and lesser COP are still challenges to the researchers. Further, environmental impacts of refrigerant is also need to be considered. Hence, suitable refrigerant with increased thermophysical properties is need to be identified for increasing the performance of commercial refrigeration system. From the literature, it is reported that the use of mixed refrigerants in refrigeration system can enhance cooling capacity. However, mixed refrigerants with nano particles are yet to be studied in commercial refrigeration applications. In this context, the thermophysical properties of nano hybrid refrigerants are investigated by using R-22 and R-12 with a mass fraction of 50% each by volume concentration of nano particle TiO₂ from 1% to 5%. Different mathematical correlations which are developed in the past are used to determine the thermophysical properties of hybrid nano refrigerants. The thermo-physical properties of hybrid refrigerants such as Specific heat, Density, Thermal conductivity and Viscosity are determined by varying temperature from 300-350K with an temperature increment of 5K. From the results, it is observed that by nano particles dispersions in mixed refrigerants increased the thermo-physical properties. Hence, it is concluded that nano hybrid refr can enhance the cooling capacity of commercial refrigeration


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Paper ID 4014

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Experimental Investigation and Optimization of wear rate in Al6062-SiC-Flyash Metal Matrix Composite

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Abstract

Aluminium Metal Matrix composites is a relatively attractive material for automobile, aerospace and other engineering application due to its mechanical and tribological properties. To improve wear resistance and mechanical properties has led to design and selection of newer variants of the composite. The present investigation deal with the study of wear behavior of Al6062-SiC-flyash MMCs for varying reinforcement content, applied load, sliding speed, and distance. Aluminium MMCs reinforced with three different percentage of reinforcement 3, 6, 9% wt. SiC and 0.5, 0.75, 1.00% wt. flyash prepared by stir casting method. Wear test was performed by using "pin on disk" apparatus. A plan of experiment based on L27 Taguchi orthogonal array is used to acquire the wear data. An analysis of variance is employed to investigate the influence of four controlling p sliding distance on dry sliding wear of the composites. It is observed that SiC content, sliding speed and normal load significantly affect the dry sliding wear. The optimal combina minimum wear. The microstructure study of worn surface indicates nature of wear to be mostly abrasive.

Keywords: Metal matrix composites; Al-SiC-flyash alloy; Stir casting; Wear; Optimization; SEM

* * *

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Experimental Investigation and Optimization of wear rate in Al6062-SiC-Flyash Metal Matrix Composite

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N.GOVINDU³

T.RAVI TEJA⁴

B.KUMAR SRINIVAS RAO⁵

Abstract

Keywords:

Metal matrix composites;
Al-SiC-flyash alloy;
Stir casting;
Wear;
Optimization;
SEM

Aluminum Metal Matrix composites is a moderately alluring material for vehicle, aviation and other designing application because of its mechanical and tribological properties. To enhance wear protection and mechanical properties has prompted outline and choice of fresher variations of the composite. The present examination manage the investigation of wear conduct of Al6062-SiC-flyash MMCs for shifting fortification substance, connected load, sliding rate, and separation. Aluminum MMCs fortified with three diverse level of support 3, 6, 9% wt. SiC and 0.5, 0.75, 1.00% wt. flyash arranged by mix throwing strategy. Wear test was performed by utilizing "stick on circle" mechanical assembly. An arrangement of analysis in view of L27 Taguchi orthogonal cluster is utilized to obtain the wear information. An examination of fluctuation is utilized to explore the impact of four controlling p sliding separation on dry sliding wear of the composites. It is watched that SiC content, sliding velocity and typical load essentially influence the dry sliding wear. The ideal combina least wear. The microstructure investigation of worn surface demonstrates nature of wear to be for the most part grating.

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Introduction

Metal Matrix Composites (MMCs) have emerged as a class of material capable of advanced properties, aerospace automotive, electronic, thermal management and wear application. The MMCs have many advantages over the conventional metal including higher specific modules, higher strength to weight ratio, better properties at elevated temperature, and lower coefficients of thermal expansion and better wear resistance. Aluminium composites are widely employed in the aerospace industry, automotive application and structural application. In the present study Al6062 composites were prepared by stir casting method reinforced with different weight % of SiC (3, 6 and 9% wt.) and flyash (0.5, 0.75 and 1.00% wt) as a reinforcement material.

Wear behavior of Al-Mg-Cu alloy reinforced with SiC particle were studied by Hassan [1] the comparative study of alloy and alloy reinforced with SiC suggested that wear resistance property of the alloy increased considerably with addition of SiC particle. Another similar study of wear behavior was performed by Kwok and Lim [2]. The composites were prepared by three different powder metallurgy techniques using different matrix metal, reinforcement weight fraction and reinforcement particle size [2]. In this work the wear studies were conducted at varying reinforcement, load, speed and distance. The wear rates are found to decrease with increasing reinforcement wt. %.

Taguchi Method

Taguchi strategy is one of the essential devices in light of performing assessment or investigations to test the affectability of an arrangement of reaction factors (autonomous or factors) by considering tests in "orthogonal exhibit" with the intend to accomplish the ideal setting of the control parameters. Orthogonal clusters give a best arrangement of very much adjusted (least) tests. Exhibit shows the quantity of line and segments it has, and furthermore the quantity of level in each of the sections. The quantity of line of an orthogonal exhibit speaks to the essential number of analyses. There are five fundamental stage, connected in Taguchi tests outline system. Stage 1-Experiment arranging, Phase 2-Design Experiment, Phase 3-Conducting Experiments, Phase 4-Analyzing Results, and Phase 5-Confirming Prediction Results.

The Taguchi technique utilizes a factual measure of performing called S/N proportion, which is logarithmic capacity of wanted yield to fill in as target work for enhancement, help in information examination and the forecast of the ideal outcomes. There are three sorts of flag to-clamor proportion of regular enthusiasm for enhancement of Static Problem; Smaller-the-better, Larger-the-better, ostensible the-best. They figure for motion to-clamor proportion are outlined so an experimenter can simply choose the biggest factor level setting to advance the quality normal for an examination. For the minimization of wear Larger-the-Better should have been utilized [3].

Paper ID 4034

Analysis Of Process Capability About Turning Center

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Abstract

Analysis of process capability convinces that processes need to be according to manufacturing industry specification, while reducing the process variation and improving the product quality characteristic. The aim is to verify the performance of the process and also the ability of the machine to achieve within the tolerance limit specified. 100 sample shafts were turned on the turning center with subgroup size being ten. The performance needs to be within the control limits and also capable of meeting up to specification. Further, the capability index (C_p) measured in this case is greater than 1. Also, the machine capability (C_{pk}) for this manufacturing application is also greater than 1, thus making the process capable.

Keywords: Turning center; Control charts; Process capability; Machine ability.

Paper ID 4035

Effect of Fiber Orientation on Strength of a Cross-Ply Composite Plate using ANSYS

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Abstract

The objective of this paper is to Superior mechanical properties of composite materials such as high stiffness and strength to weight ratios, corrosive resistance and low coefficients of thermal expansion caused it to be used increasingly in many areas of technology including marine, aerospace, automobiles and others. To take advantage of full potential of composite materials, accurate models and design method are required the most common structural elements are plates and shells. Laminated composites are one of the classifications of the composites which are used in structural elements like leaf springs, automobile drive shafts, and gears, and axles. An accurate understanding of their structural behaviour is required such as the deflections, the through thickness distributions of stresses and strains, the large deflection behaviour and, of extreme importance for obtaining strong, reliable multi-layered structures. In general, the fibres in the composite materials can be of unidirectional and bidirectional. Here the focus is on unidirectional fibre composite materials. The fibres can be oriented in many ways i.e. 0, 90, 20, 30, 45, 90 degrees etc., but the strength in each type will be different. Using the load-deflection graphs, the maximum load and flexural strength are calculated. The attention is to note the variation of flexural strength of a composite material with the variation of fibre orientation and comparing the results with practical results. Here the modelling of composite material with different fibre orientation is done in Ansys software and simulation is done on them and those results are compared with practical results.

Keywords: Flexural strength; Laminated; Fibre reinforced; Glass fibre; Graphite fibre; Unidirectional; Bidirectional; Corrosion resistance.

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Keywords: Flexural strength; Laminated; Fibre reinforced; Glass fibre; Graphite fibre; Unidirectional; Bidirectional; Corrosion resistance.

2nd International Conference on Mathematical Sciences in Engineering Applications (ICMSEA - 2017)

Paper ID 4001

Design and Methodology of Line Follower Automated Guided Vehicle for Material Handling

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Abstract

Advancement of automated/computerized guided vehicle assumes a noteworthy part in designing businesses to enhance the material taking care of procedure for late year. In this paper, it is focus on the design and different methodology of line follower automated guided vehicle (AGV) systems. This paper provides an overview on line follower AGV discusses recent technological developments. The essential components of line follower robot and their modification are described in this paper.

Keywords: Material handling; Automation; Line follower agv; AGV.


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Paper ID 4031

Fabrication and Experimental Investigation on Mechanical Properties of Aluminium Metal Matrix Composite by using Stir Casting

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Abstract

From the last two decades there is a rapid increase in utilisation of Aluminium Alloy, particularly in Automobile, Aerospace industries, due to its low density, low weight and high strength. Aluminium metal matrix composite has received extensive attention for practical as well as fundamental reasons. Aluminium Alloy & Aluminium based metal matrix composites (MMC) have found many applications in the manufacturing of various components. The present work is focused on Fabrication & Investigation of Mechanical properties of Al6061, Graphite, Fly ash metal matrix composite. These materials are identified in the process of making high strength Aluminium Alloy. Stir casting process is employed for the fabrication of metal matrix composite. In our project the composition of MMC is varied for graphite (%, 6%) & fly ash (6%, 3%) by keeping 91% by Al 6061. Fabricated samples are tested for Mechanical properties. The Tests includes Tensile test, Hardness test, and Wear test as per ASTM standards. Microstructures of the samples were tested by using metallurgical microscope & compare the microstructure for each.

Keywords: Composite materials; Mechanical properties; Metal matrix composites; Stir casting; Micro structure.

Design and Analysis of Aircraft Wing using Composite Materials for varying Angle of Attack

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Abstract

In present scenario, the design of Aircraft became challenging due to problem that are encountered in engineering practice. Different analysis are done in the past to develop a better Aircraft. However, there are many parameters that influences the Aircraft design are drag, lift, static and dynamic pressure, winglets design, angle of attack, vortex developed, gravity and angle of flight. Hence, the development in actual practice are not efficient and prone to accidents. In this context, an aircraft wing is designed by changing the material for finding better material suitable. The 3D model of NACA 4412 profile is designed in CATIA V5 and exported to commercial Ansys 14.5 Software. By applying the boundary conditions similar to the working environment the analysis is carried. The structural analysis of aircraft wing made of materials such as AA7075 and a composite GLARE (glass-reinforced aluminum laminate) is done under steady inertia loading. Modal analysis also done for defining the natural frequency and mode shape of continuous structures. Further, by varying the angle of attack (AOA) from 00 to 80 for an inlet velocity of 10m/s the static and dynamic pressure of aerofoil and pressure coefficients are determined using Computational Fluid Dynamics (CFD). The pressure based steady state solver with K- ϵ turbulence scheme is used for determining the pressures and their coefficients. The results of the analysis are stated in the literature.

Keywords: Air craft wing; Composite materials; Angle of attack; Computational Fluid Dynamics; Glass reinforced composites.



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MallikarjunaPotta****

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SHAURIDUT- The solar car of BITS Vizag

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Abstract

In urban communities of India one of the real medium of transportation is auto rickshaws, which delivering a colossal measure of air contamination and also nursery gasses like CO₂. Fuel, which utilized is a non-inexhaustible source and furthermore which costs, one of the front runners within vicinity of renewable energy resources nowadays is sun strength. a solid source like sun oriented vit which is accessible in bounty in a nation like India. Received SOLAR ENERGY as the extra sou notwithstanding the regular IC Motors. Utilizing the Photovoltaic panels, controllers, and brushless motor setup to change over the light, as an electric vitality which is sustained to the DC engine to act

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SHAURIDUT- The solar car of BITS Vizag

HEPSIBA SEELI*

VIKAS RANJAN**

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Abstract

In urban communities of India one of the real medium of transportation is auto rickshaws, which is delivering a colossal measure of air contamination and also nursery gasses like CO₂. Fuel, which is utilized is a non-inexhaustible source and furthermore which costs, one of the front runners within the vicinity of renewable energy resources nowadays is sun strength, a solid source like sun oriented vitality which is accessible in bounty in a nation like India. Received SOLAR ENERGY as the extra sources notwithstanding the regular IC Motors. Utilizing the Photovoltaic panels, controllers, and brushless DC motor setup to change over the light, as an electric vitality which is sustained to the DC engine to acquire mechanical movement. The mechanical movement was exchanged to the wheels through the propeller shaft which prompts cheap and powerful transmission.

The paper represents how the charge produced by a variety of sun-powered boards are gotten and its stream all through a battery pack is to be controlled utilizing a microcontroller based charge controller to guarantee effective putting away of charge in a battery pack. The putaway vitality would be unveiled to a DC engine which would run the auto. The design of an engine controller to control the auto's speed and forward/turn around bearing of movement appears. The mechanical development from scratch of the chassis along with all crucial mechanical frameworks of Shauridut solar car is illustrated.

Keywords:

Solar car, Solar Energy,
Photovoltaic panels, Brushless DC
motor, CONTROLLER,
Batteries,

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Paper ID 4040

Numerical And Experimental Analysis Of Thermal Performance In A Parallel Flow And Counter Flow Double Pipe Heat Exchange

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Abstract

Heat transfer is taken into consideration as transfer of thermal energy from bodily frame to another. It is the most essential parameter to be measured because the overall performance and efficiency of the double pipe heat exchanger. By the use of CFD simulation software, it reduces the time and operation cost in comparison to experimental calculations. The goal of this paper is to evaluate the influence of the Double pipe heat exchanger to get better heat transfer. The purpose of this examine is to apply CFD software program and Experimental setup to analyze the Temperature drop, Pressure drop and Friction factors, by varying under different inlet conditions like Temperature and Flow rate as a function of both inlet velocity and temperature variations and converting heat exchanger tube material properties like copper and aluminium. In this experiment, heat transfer from hot fluid to cold fluid with the aid of Double pipe heat exchanger is experimentally by using different organic fluids like Benzene, Glycol, Transformer oil, Acetone and Water to get better heat transfer, the identical thing is established in CFD analysis. The test is accomplished at Laminar flow beneath different flow preparations like Parallel flow and Counterflow. The test is done on double pipe heat exchanger under various working fluids with different operating conditions, it is predicted that Counterflow exhibits better heat transfer than Parallel flow.

Keywords: Heat transfer, Double pipe heat exchanger, CFD simulation software, Organic fluids, Copper, Aluminium, Laminar flow, Turbulent flow, Parallel flow and Counterflow

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Experimental Data of Calophyllum Inophyllum Bio Diesel Effect by Varying Compression Ratio and Injection Pressure on Emissions

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Abstract

Degrading fossil fuel resources have thrown world into chaos, besides air pollution and green house gases are a major concern. The fossil fuel shortage in near future is inevitable. This situation triggered the awareness to find alternative and sustainable energy resources. Biodiesel is a better solution as its properties are close to petrodiesel. In this paper, Biodiesel is produced from Calophyllum inophyllum oil by two stage transesterification process using acid and base catalysts. Acid esterification is done by mixing 1% (by wt. of oil) of 99% pure anhydrous sulphuric acid in 1000 ml of oil with oil to methanol ratio 6 and maintained at a temperature of 55°C for 120 minutes. This reduces the acid value of the oil. Next the oil is subjected to base catalyst. 1% of KOH (by wt. of oil) is added to a mixture of oil to methanol ratio 4:1. The mixture is stirred for two hours at 60°C. The yield is about 89%. Due to high viscosity is difficult for regular engines to inject biodiesel into combustion chamber. This results in improper mixing of fuel and reduces ignition quality. In this paper the fuel injection pressure has been increased and compression ratio has been varied and tested under different operating conditions. The increase in injection pressure resulted in better atomization of fuel and better mixing with the air in combustion chamber. The increase in compression ratio resulted in higher compression pressure which ensures complete combustion of fuel in the chamber. These modifications result in reduction of emissions and smoke to about 10% compared to normal engines running on biodiesel.

Keywords: Biodiesel, Transesterification, VCR Engine, Injection pressure, compression ratio, Engine emission.



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Fabrication and Experimental Investigation on Mechanical Properties of Aluminium Metal Matrix Composite by using Stir Casting

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Abstract

From the last two decades there is a rapid increase in utilisation of Aluminium Alloy, particularly Automobile, Aerospace industries, due to its low density, low weight and high strength. Aluminium metal matrix composite has received extensive attention for practical as well as fundamental research. Aluminium Alloy & Aluminium based metal matrix composites (MMC) have found many applications.

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Paper ID 4033

Reduction Of Bio Diesel Emissions By Injection Advance In VCR Engine

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Abstract

Biodiesel in present trend is important due to depletion of non-renewable resources. The shortage of fossil fuels brings usage of bio diesel into limelight. Therefore, usage of bio diesel is preferred than usage of conventional diesel as the properties of both are almost same. A lot of research has undergone on bio diesel and doesn't show any significant variation in emissions compared to conventional diesel. Due to high emissions of harmful gases global warming increases which is not preferable. In this paper Flax seed is used for bio diesel preparation which is abundantly available in mountainous areas and cooler regions. Flax seed commonly known as lin seed has binomial name *Linum usitatissimum*. In this process the oil is transesterified using methanol to produce methyl esters in presence of base catalyst. Methanol is mixed with flax seed oil in the ratio 1:5 and 0.5 to 5% base catalyst is added. This mixture is constantly stirred at 55°C. Due to high viscosity it is difficult for biodiesel to mix with air in the combustion chamber and results in increase of emissions. As the fuel injection is advanced by change in crank angle up to certain extent in a VCR engine, the time available for mixing fuel with air is increased ensuring increased mixing of fuel with air, and By increasing the compression ratio the pressure in combustion chamber increases which helps high viscous fuel to mix completely, due to high temperature because of the high pressure in the combustion chamber the combustion characteristics of charge increased results in lowering the engine emissions. The above tests were conducted using conventional diesel, and different proportions of blends of bio diesel and results were compared by varying crank angle and compression ratios.

Keywords: Biodiesel, Transesterification, Crank angle, Emissions



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The Role Of Women Entrepreneurs As A Change Agent In The Society: A Case Study

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MR P.B. Narendra Kumar, HOD, MBA Department, BABA Institute of Technology and Sciences (BITS), PM Palem, Visakhapatnam- 500048

Abstract

A society that does not have an optimistic, positive empowering image of the future is endangered. When society is skewed & power, it begins an entropic spiral to isolation – social entrepreneurs can use that skills to rebalance society. Society entrepreneurs are change agents that improve society by developing effective and equitable new models often less hierarchical yet more cooperative and complex than existing ones. Researchers and practitioners have defined society entrepreneurship in various ways but a common demonstrator is a venture that adds value to a community mission through innovative, risk-taking, business-like practices. Society entrepreneurship combines innovation with the importance of community needs and their work overlaps with social justice and environmental presentation movements. Women in Ancient India were in charge of the home affairs and were accorded very high status almost equal to that of man in religious duties. Gradually due to the varying social-political situation, women were relegated to the background and were subjected to exploitation.

Keywords: empowerment, entrepreneur, innovation.

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Paper ID:7009

Training And Development Practices In Select Corporate Retail Stores In Visakhapatnam

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Abstract

A profound training program acts as a vehicle to enhance employee skills and enable them to perform better in their job. Training and development is very crucial to the employees, the organization and their effectiveness. Training is important in retailing because majority of the employees have direct contact with customers. They are responsible for helping customers, satisfy their needs and resolve their problems. Training is important not only from the point of view of employees, but also for the organization. This study has examined the perception of respondents on the present training and development practices of select corporate retail organizations. With this background, this study is conducted to know the perception of respondents in terms of their demographic profiles of gender, age, education and social status.

Keywords: Training, Development, Promotion


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Paper ID:7013

Operational And Financial Performance of Urban Cooperative Banks In India

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Mr. P. Manoj Babu, Asst. Professor, Department of Management Studies, BITS Visakhapatnam

Abstract

This study attempts to measure the operational and financial performance of Urban Cooperative Banks in India. The analytical frame of the operational tools in terms of the Credit Deposit Ratio and Investment Deposit Ratio, and two different financial segments, purposes; and also, the trends over the period under study have been presented. The study is based on secondary data collected from annual reports of urban cooperative banks in India for the study period i.e. 2005-06 to 2015-16. Ratios are used here with financial ratio analysis (FRA) method which helps to draw an overview about the financial performance of the urban cooperative banks in terms of profitability, liquidity and credit performance. To test the hypothesis the study has been worked on Statistical tools, correlation, t-test by using SPSS. These analyses are helped to see the current performance, condition of these urban banks to compare past performance. The study findings can be helpful for managing of urban cooperative banks. The study also identified specific areas for urban cooperative banks to work on which can ensure sustainable growth and development for these scheduled banks and non-scheduled banks.

Keywords- Operational Performance, Profitability, Financial ratio analysis, Credit Deposits Ratio and Investment Ratio.



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Mr. P. Manoj Babu, Asst. Professor, Department of Management Studies, BITS Visakhapatnam

Abstract

This study attempts to measure the operational and financial performance of Urban Cooperative Banks in India. The analytical frame of the operational tools in terms of the Credit Deposit Ratio and Investment Deposit Ratio, and two different financial segments, purposes; and also, the trends over the period under study have been presented. The study is based on secondary data collected from annual reports of urban cooperative banks in India for the study period i.e. 2005-06 to 2015-16. Ratios are used here with financial ratio analysis (FRA) method which helps to draw an overview about the financial performance of the urban cooperative banks in terms of profitability, liquidity and credit performance. To test the hypothesis the study has been worked on Statistical tools, correlation, t-test by using SPSS. These analyses are helped to see the current performance, condition of these urban banks to compare past performance.

The study findings can be helpful for managing of urban cooperative banks. The study also identified specific areas for urban cooperative banks to work on which can ensure sustainable growth and development for these scheduled banks and non-scheduled banks.

Keywords- Operational Performance, Profitability, Financial ratio analysis, Credit Deposits Ratio and Investment Ratio.



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Productivity Performance of The Indian Foreign Banks: An Experimental Analysis

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Abstract

Indian banking system has transformed in recent years due to globalization in the world market, which has resulted in fierce competition. In this paper, an attempt has been made to analyze the productivity performance of Foreign Banks in India. Analytical frame of the operational performance per employee in terms of various tools Deposit, Advances, Total business, Total expenses, Establishment, Net profit and Investment, and to different financial segments, purposes; and also the trends over the period under study have been presented. The study is based on secondary data collected from annual reports of five selected foreign banks in India for the study period i.e. 2004-05 to 2015-16. Ratios, t-test, correlation analysis have been used to analyze the present data. The study suggested that foreign banks in India should emphasize on generating more profits per employee by efficient utilization of its capital, total investments, advances, deposits, debt and improving its operational efficiency and productivity performance.

Keywords: Total Business, Deposits per employee Investment per employee, productivity performance


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A Comparative Study on Work-Life Balance between Married and Unmarried Female Teaching Fraternity in an Engineering College

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Abstract

Work life balance plays a dominant role in an individual life and it was a challenging task especially for women to manage both the work life and personal life in this modern era. The present research study identifies the causes of imbalance of work and personal life among the married and unmarried female teaching fraternity in selected engineering college. This paper analyzes the causes of work and life imbalance with reference to female faculty. A total of 52 faculty responses of a selected engineering college are included in the study. Statistical Analysis reveals that cause of imbalance of work life balance is spending more hours to travel to reach the work place, carrying work to home, not having enough time


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VISAKHAPATNAM - 530 048

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Paper ID:7026

Consumer Attitude Towards The Products Of Hyper Markets In Visakhapatnam

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BITS VIZAG bramajyothi@bitsvizag.com; vasumalicherla@gmail.com; nareshpatnaik.k@gmail.com

Abstract

Market is a place where buyers and sellers meet, goods and services are offered for sale and transfer of ownership occurs. Sometimes a market is a precisely physical place and sometimes it refers to groups of people. At present day's hyper markets plays an important role to develop the modern markets and improve the customer services and promote the economic excellaration. As market consists of consumers, their attitude, beliefs, loyalty and perception is important in modern marketing. Without customers, a firm has no revenues, no profits, and therefore no market value. Consumers all have different tastes, like and dislikes and adopt different behaviour patterns while making purchase decisions. The present study deals with the attitude, consumer satisfaction and preferences with regard to the services, prices, quality, products etc. of hyper markets.

Keywords:Modern Markets; Customer Attitude; Customer satisfaction; Hyper markets;

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Paper ID:7026

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Keywords:Modern Markets; Customer Attitude; Customer satisfaction; Hyper markets;

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First Year Student Expectations Of An Undergraduate Management Program: A Survey

Dr. G N P V Babu, Associate Professor, Baba Institute of Technology and Sciences, Visakhapatnam, Andhra Pradesh/

Mr. Rajesh Dorbala, Assistant Professor, Mittal School of Business, Lovely Professional University, Phagwara, Punjab

Abstract

Although much has been written on the first-year experience of students at higher education institutions, less attention has been directed to the expectations of students when they enter an institution for the first time. This paper delivers acumens into the expectations of first year students at largest private university in India and highlight areas in which students' expectations may not essentially line up with the realities of common university practices. By providing chances for students to coherent their expectations, staff are able to use the responses for a fruitful dialogue and work towards a more positive alignment between perceived expectations and levels of student satisfaction with their experience. It highlights that the expectations of the students are beyond traditional teaching methods and it is inclined towards career planning and orientation.

Keywords: Student Perception, Career Orientation, Traditional Teaching, Skill Set

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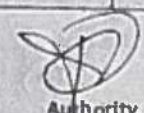
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ANALYSIS OF PROCESS CAPABILITY ABOUT TURNING CENTER

Chundru.Venkata Rao¹

N.Tejeswar Reddy¹

B.Sai Prasanth¹

Pujari Srinivasa Rao²

P.Thirumala Rao¹

Abstract

Analysis of process capability convinces that processes need to be according to manufacturing industry specification, while reducing the process variation and improving the product quality characteristic. The aim is to verify the performance of the process and also the ability of the machine to achieve within the tolerance limit specified. 100 sample shafts were turned on the turning center with subgroup size being ten. The performance needs to be within the control limits and also capable of meeting up to specification. Further, the capability index (Cp) measured in this case is greater than 1. Also, the machine capability (Cpk) for this manufacturing application is also greater than 1, thus making the process capable.

Keywords:

Turning center;
Control charts;
Process capability;
Machine ability.

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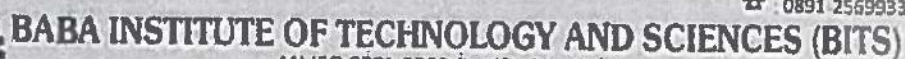
1. Introduction

Nowadays, most of the manufacturing companies spend a lot of money for machine tools with advanced technology such as CNC tools and automated machine tools. These will help to improve the quality and dispatch time of the products that they produced. The investors require that the machines they purchase have high flexibility, good running condition and highly precision-oriented turning centers. In the manufacturing sector, the accuracy of parts is very important because the accuracy of the finish part helps in evaluating the machine's accuracy that produces the particular part. In the early 1970s the concept of process capability was introduced by Juran, an expert in world quality. Subsequently, for almost 40 years many accepted indices such as Cp, Cpk and Cpm were developed to measure the performance of the respective process [1]. The measure of the uniformity of performance is process capability. Variability of the process output is measured by its uniformity in process product. The relative deviation in the product specification and requirements is nothing but variability in process product. Process capability analysis is a technique applied in many stages of the product cycle. This includes product design, manufacturing planning process and production in machines for each operation. This technique helps in determining the ability for achieving the engineering values and manufacturing parts within the tolerance limits.

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The long term performance level of a process which is brought under statistical control is called process capability. Statistical process control is a quality assurance tool which is used to improve the quality of manufacture and helps in scoring the end-customer satisfaction [2]. The ability of the combination of the



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Design and fabrication of Three Axes Pneumatic Trailer for shipping Goods

Dharmala Venkata Padmaja*

Yalagada Aravind Kumar**

Kallepalli Rambabu***

Pradhan Hemanta Kumar****

Jami Rajesh Babu*****

Mandangi Dilip Kumar*****

Abstract

The objective of this paper is to evaluate the difficulty in unloading the materials. In this consideration, several automobiles garages have concede the facts that, some of the methods, which were most difficult, were endorsed in unloading the materials from the trailer. The loading and unloading of material using trailer is only in one single direction. Moreover, loading and unloading of material, in small compact streets and small roads, is done with high difficulty.

Now our project has mainly concentrated to rectify the loading and unloading of the trailer in all the three sides very easily with suitable arrangement of design. The loading of vehicle from the trailer can be done in three axes by activating the direction control valve, without the application of any impact forces. Compressed air is used in controlling the valve. The trailer cabin is lifted, by using the ram of the pneumatic cylinder as lifter. The compressor engine is coupled with automobile engine drive, so that, when vehicle is running, compressed air is stored. Pneumatic cylinder is activated by using the compressed air, when the valve is activated.

Keywords:

Trailers
Control Valve
Pneumatic Cylinder
Compressed air

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1. Introduction

In building sites, the vehicle used in carry materials is dumper. Dumpers are differentiated from dump trucks by structure; dump truck is having a cab in front of the load, while an open 4-wheeled vehicle with the load skip in front of the driver is a dumper. The name dumper has come from "The skip can tip to dump the load". They are usually powered with diesel. For the secondary use as a site tractor, a towing eye is fitted.



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DESIGN AND FABRICATION OF INDUSTRIAL PARAMETER MONITORING AND CONTROL ROBOT

CH Uma Chaithanya*

K Sri Ram Murthy**

P Silva Vasu***

D Rama Krushna****

K Yuva Kishore*****

K P Dala Krishna*****

Abstract

Now-a-days the accidents in the coal mine industries have increased. Even if any explosion occurs it can't be easily known to the worker and it may cause accidents. So in order to avoid this, a robot has been designed and this robot is allowed to monitor the ambient situations inside the coal mine industry. Some of the environmental parameters such as methane leakage, temperature, oxygen are sensed by using the high end sensors and the sensed data are transmitted for monitoring the status of the coal mine and to control the robot movement.

Keywords:

DC Motors;
Sensors;
Buzzer;
Battery;

If the temperature exceeds a threshold, the cooling fan is automatically set to ON and if any gas leakage is detected the workers are given alert through a buzzer. By this the human intervention can be avoided inside the industry and the accidents can be prevented.

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